Anyu Pan

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EDUCATION

University of Pennsylvania, USA 2024-2027

- M.S. (expected) in Computer and Information Science
- M.S. (expected) in Bioengineering

University of Dundee, UK 2020-2024

Joint Undergraduate Programme with Northeastern University , China

- B.Eng. in Biomedical Engineering (First Class Honours)

SKILLS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, MATLAB, R, Dart

ML/DL & Tools: PyTorch, Scikit-Learn, Deep Learning (Transformers/CNNs), AWS (EC2, S3), Git

SELECTED PROJECTS

Internet and Web Systems: Distributed Search Engine | Member 11/2025-12/2025

- Hardened **crawler** with **socket/HTTP timeouts** + robots handling; tuned GET/HEAD timeouts to crawl **100K+ pages in ~9 hrs.**
- Optimized distributed **PageRank** via **per-partition KVS writes** and reduced client churn; fixed **KVS persistence** and **Flame ring-ID partitioning.**
- Improved search UX with **autocomplete**; normalized result URLs and added **title/snippet fallbacks**; deployed/monitored jobs on **AWS EC2** and benchmarked instance types.

Advancing Glioma Classification with Swin Transformers | Honor Project | University of Dundee & University of Cambridge 09/2023-05/2024

- Integrated **Swin Transformer blocks** into a deep learning WSI-based glioma classification framework, improving accuracy to 78.03% (from 73.11% with a standard ViT)
- Systematically evaluated multiple **optimizers** (SGD, Momentum, AdaGrad, RMSprop, Adam, and Lion) to identify optimal performance-efficiency trade-offs
- Reduced GPU memory usage by **>60%**, enabling larger batch sizes and more efficient training on GPU-accelerated servers

Brain–Computer Interface: Decoding Finger Flexion from ECoG | Member 03/2025-05/2025

- Collaboratively developed a **convolutional encoder–decoder** to predict five-finger flexion trajectories from **ECoG signals** (BCI Competition IV dataset)
- Implemented signal preprocessing and feature extraction pipeline (bandpass/notch filtering, **Morlet wavelet convolution,** downsampling, lag correction) to prepare high-frequency neural data
- Achieved mean Pearson correlation of 0.624 on the held-out test set using a combined **MSE + cosine loss**

PUBLICATIONS

- Qiu, Z., **Pan, A.**, & Qi, L. (2023). Incorporating dynamic attention gating mechanism and pre-trained embedding for Chinese clinical named entity recognition. *Procedia Computer Science, 226*, 113-119.

AWARDS & HONORS

- **2023 ASA Data Insights Award(top 0.1%),** Mathematical Contest In Modeling, International Level 11/2023
- **Outstanding Winner (top 0.2%) & COMAP Scholarship Award & ASA Award,** Mathematical Contest In Modeling, International Level 05/2023

PART-TIME EXPERIENCE

Developmental Engineering Of Tissues | Teaching Assistant | Penn Engineering 08/2024-12/2024